

2. Control Statements

- ♦ Control Statements are used to control the execution flow of Java Program.

There are three types of control statements:

1) Conditional Control Statements

- if statement
- switch statement

2) Looping Control Statements

- for statement
- Enhanced for Statement (Added in Java 5)
- while statement
- do-while statement

3) Unconditional Control Statements

- break
- continue

2.1. Conditional Control Statements

- ♦ Conditional Control Statements are also called as decision-making statements or selection statements.
- ♦ It allows the program to select between alternative actions during the program execution.
- ♦ The execution will be based on the result of conditional expression.

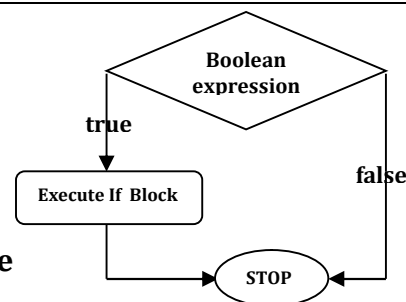
2.1.1. if Statement

1) Simple If

```
if(<condition>){  
    // Statements (IF BLOCK)  
}
```

```
// Other Statements
```

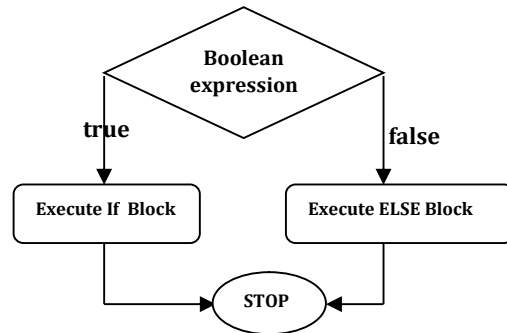
- ♦ It indicates whether IF BLOCK Statements will be executed or not, as per the condition.





2) If-else Statement

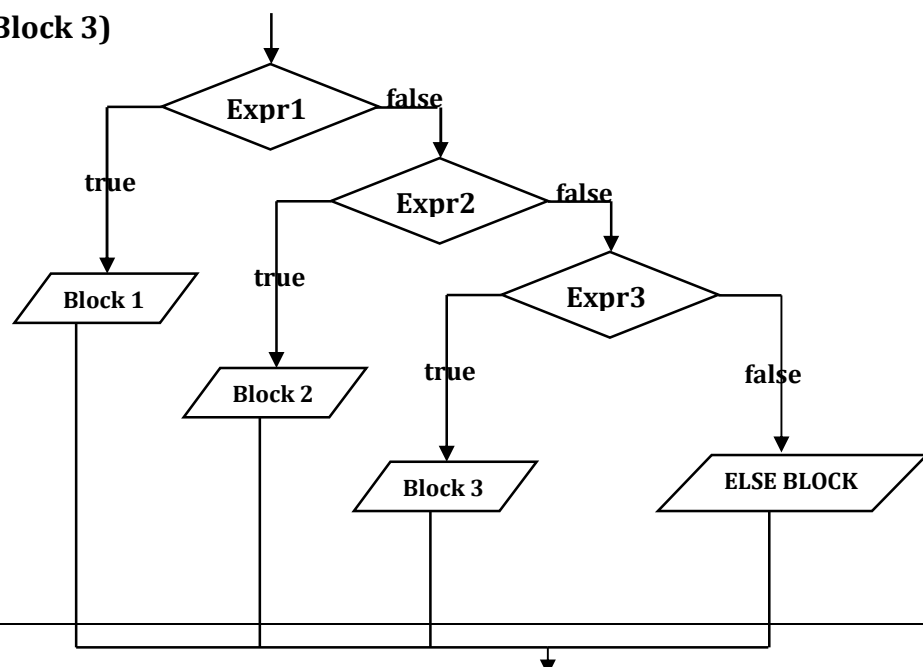
```
if(<condition>){  
    // Statements (IF BLOCK)  
}else{  
    // Statements (ELSE BLOCK)  
}
```



- ♦ It indicates which block will be executed, as per the condition.
- ♦ if condition is true IF BLOCK Statements will be executed otherwise ELSE BLOCK Statements will be executed.

3) If Else-if statement

```
if(<condition1>){  
    // Statements (Block 1)  
}else if(<condition2>){  
    // Statements (Block 2)  
}else if(<condition3>){  
    // Statements (Block 3)  
}  
...  
else{  
    // Statements  
}
```



- ♦ Condition can be any expression which returns boolean value.

- ♦ **Block notation is mandatory when if block or else block have more than one statement.**
- ♦ **Block notation is optional when if block or else block have only one statement.**
- ♦ **When you declare any variable within if block or else block then that variable can be accessed within that block only. It can't be accessed from outside that block.**

1) Demo1.java

```
package com.myjlc.demos;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
class Demo1{
    public static void main(String as[]) {
        int a=42;
        if(a % 2==0)
            System.out.println("Number is EVEN");
        else
            System.out.println("Number is ODD");
    }
}
```

2) Demo2.java

```
package com.myjlc.demos;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
class Demo2{
    public static void main(String as[]) {
        int a=45;
        if(a % 5==0)
            System.out.println("Number is Divisible by 5");
        else
            System.out.println("Number is Not Divisible by 5");
    }
}
```



3) Demo3.java

```
package com.myjlc.demos;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

class Demo3{
public static void main(String as[]) {
int a=10;
int b=20;
int max=0;
if(a>b)
    max=a;
else
    max = b;

System.out.println("Max is "+max);

}
}
```

4) Demo4.java

```
package com.myjlc.demos;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

class Demo4{
public static void main(String as[]) {
int a=10;
int b=20;
int c=15;
int max=0;
if(a>b && a> c)
    max=a;
else if(b> a && b>c)
    max = b;
}
```

```
else
    max=c;
System.out.println("Max is "+max);
}
}
```

2.1.2 switch Statement

- ♦ Switch statement allows the program to select one action among multiple actions during the program execution.

Syntax:

```
switch(<variable/expression/value>){
    case <val1>:
        // Statements    [<break>];
    case <val2>:
        // Statements    [<break>];
    ...
    default:
        // Statements
}
```

- ♦ First switch expression will be evaluated and then tries to find case value matching with resultant value of the expression.
- ♦ If matching case is found then the statements related to that case will be executed otherwise statements related to default case will be executed.
- ♦ Following types of values can be passed in switch as an argument up to Java 2:
 - byte
 - short
 - char
 - int
- ♦ enum can be passed in switch as an argument from Java 5.
- ♦ String can also be passed in switch as an argument from Java 7.



5) Demo5.java

```
package com.myjlc.demos;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
class Demo5{
public static void main(String as[]) {
int a=3;
switch(a){
case 0:
    System.out.println("Sunday");
    break;
case 1:
    System.out.println("Monday");
    break;
case 2:
    System.out.println("Tuesday");
    break;
case 3:
    System.out.println("Wednesday");
    break;
case 4:
    System.out.println("Thursday");
    break;
case 5:
    System.out.println("Friday");
    break;
case 6:
    System.out.println("Saturday");
    break;
default:
    System.out.println("Wrong Input");
}
}
}
```



6) Demo6.java

```
package com.myjlc.demos;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
class Demo5{
public static void main(String as[]) {

char ch='A';
if(ch>=65 && ch<=90){
switch(ch){
case 'A':
case 'E':
case 'I':
case 'O':
case 'U':
    System.out.println("Character is Vowel");
    break;
default:
    System.out.println("Character is Consonant");
}
}else{
    System.out.println("Invalid Alphabet");
}

}
}
```



2.2. Looping Control Statements

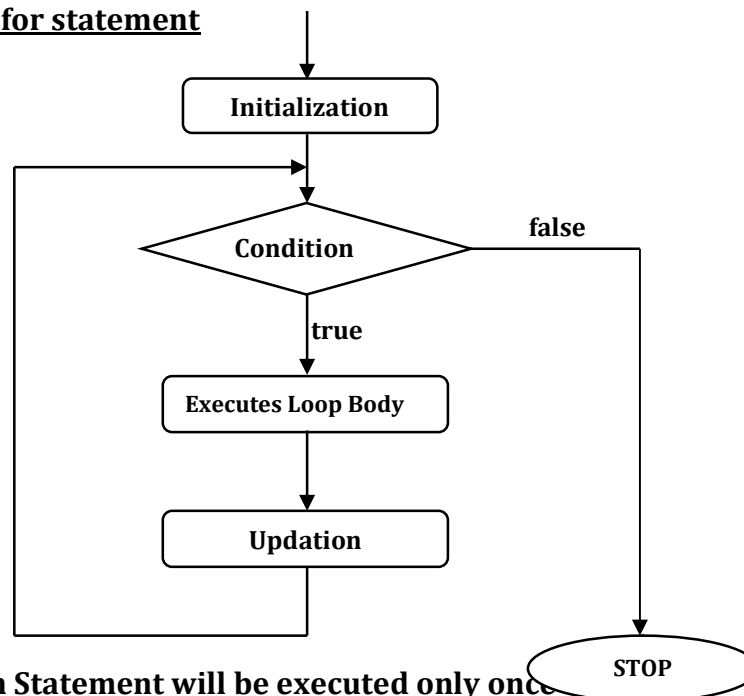
- ♦ Looping Control statements are also called as Iteration statements.
- ♦ These statements are used to execute a block of statements repeatedly.
- ♦ An expression is required for looping control statements which returns boolean value.
- ♦ Block notation is mandatory when looping block has more than one statement.
- ♦ Block notation is optional when looping block has only one statement.
- ♦ When you declare any variable within loop block then that variable can be accessed within that block only. It can't be accessed from outside that block.

2.2.1 for Statement

Syntax:

```
for(<initialization> ; <condition> ; <updation>)  
{  
  // Statements  
}
```

Processing Flow of for statement



- Initialization Statement will be executed only once.
- Conditional & Updation statement can be executed more than once.



7) Demo7.java

```
package com.myjlc.demos;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

class Demo7{
public static void main(String as[]) {

int num=10;
for(int i=1;i<=num;i++){
System.out.println(i);
}

}
}
```

8) Demo8.java

```
package com.myjlc.demos;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

class Demo8{
public static void main(String as[]) {

int num=25;
for(int i=1;i<=num;i++){
if(i % 2 == 0)
System.out.println(i);
}
}
}
```

2.2.2 Enhanced For Statement

- ♦ It is a new feature added in Java 5.
- ♦ It is also called as for-each loop.
- ♦ The enhanced for-loop is mainly used to access the elements from Collection/Array sequentially.

Syntax:

```
for(<data Type> varName : array | Collection) {  
    // Statements (BLOCK)  
}
```

9) Demo9.java

```
package com.jlccindia;  
  
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */  
class Demo9{  
    public static void main(String as[]) {  
  
        int arr[]={99,88,77,66,55};  
  
        System.out.println("Length :"+arr.length);  
  
        for(int element []: arr){  
            System.out.println(element);  
        }  
    }  
}
```



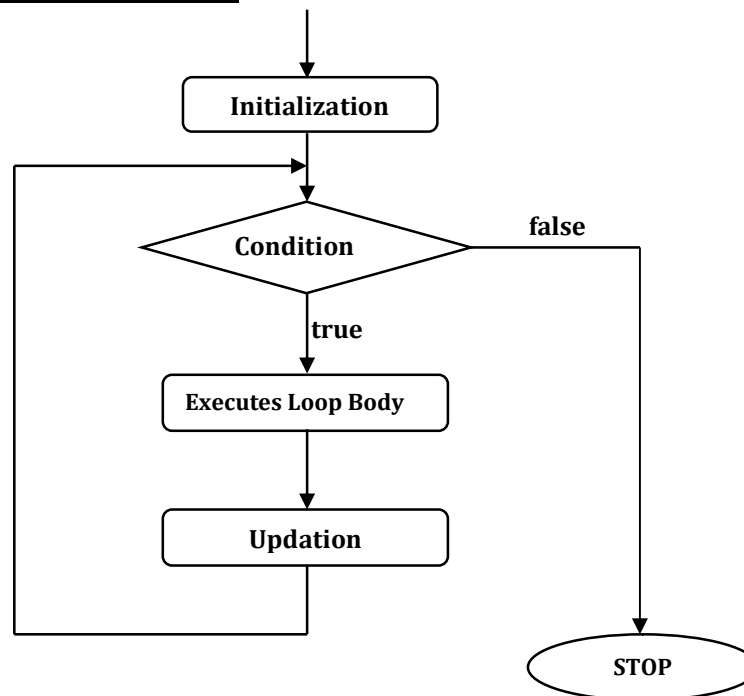
2.2.3 while Statement

Syntax:

```
<initialization>
while( <condition>){
    // Statements (LOOP BODY)
<updation>
}
```

- ♦ condition of while statement is mandatory and must be boolean type.

Processing Flow of while statement



10) Demo10.java

```
package com.myjlc.demos;
class Demo10{
public static void main(String as[]) {
int i=1;
while(i<=10){
System.out.println(i);
i++;
}
```

```
}  
}
```

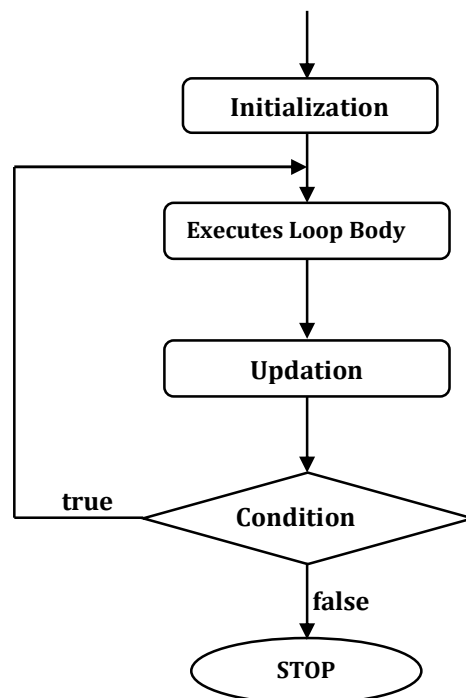
2.2.4 do-while Statement

- ♦ When you are using for statement or while statement then it verifies the condition before executing the block.
- ♦ So in the case of for statement and while statement, when first time condition is false then the block will not be executed.
- ♦ for and while statement are also called as Entry Controlled Loop.
- ♦ If you want to execute the block at least once then use do-while statement.

Using do-while loop

- ♦ In the case of do-while first block of statements will be executed and then condition will be verified.
- ♦ do-while statement is also called as Exit Controlled Loop.

Processing Flow of while statement





11) Demo11.java

```
package com.jlcindia;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
class Demo11{
public static void main(String as[]) {

int i=1;
do{
System.out.println(i);
i++;
}while(i<=10);

}
}
```

2.3 Unconditional Control Statements

2.3.1 break Statement

- ◆ break is a keyword.
- ◆ It can be used within switch or any looping statements.
- ◆ It is used to terminate the execution of the current looping/switch statement.
- ◆ break can be used in two ways:
 - break;
 - break <label>;

12) Demo12.java

```
package com.jlcindia;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
```



```
class Demo12{
public static void main(String as[]) {

for(int i=0;i<10;i++){
System.out.println(i);
break;
System.out.println("---Inside Loop---");
}
System.out.println("---Outside Loop---");

}
}
```

2.3.2 continue Statement

- ♦ **continue** is a keyword.
- ♦ It can be used within any looping statements.
- ♦ It is used to continue the execution of the current looping statement with next iteration.
- ♦ **continue** can be used in two ways:
 - **continue;**
 - **continue <label>;**

13) Demo13.java

```
package com.jlcindia;

class Demo13{
public static void main(String as[]) {

for(int i=0;i<10;i++){
System.out.println(i);
continue;
System.out.println("---Inside Loop---");
}
System.out.println("---Outside Loop---");

}
}
```



2.4. Programs on Numbers and Loops

1) Lab1.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

// 1. Swap 2 Numbers.

public class Lab1 {

    public static void main(String[] args) {
        int a=10;
        int b=20;

        System.out.println(a+"\t"+b);

        int temp= a;
        a=b;
        b=temp;

        System.out.println(a+"\t"+b);

    }

}

// Time Complexity - O(1)
//Space Complexity - O(1)
```



2) Lab2.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 * */
```

```
//2. Swap 2 Numbers without 3rd Variable
```

```
public class Lab2 {  
  
    public static void main(String[] args) {  
        int a=10;  
        int b=20;  
  
        System.out.println(a+"\t"+b);  
  
        a=a+b;  
        b=a-b;  
        a=a-b;  
  
        System.out.println(a+"\t"+b);  
    }  
}
```

```
// Time Complexity - O(1)
```

```
// Space Complexity - O(1)
```

```
//This Logic Works only for Numbers.
```




3) Lab3.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

//3. Swap 2 Strings

```
public class Lab3 {  
  
    public static void main(String[] args) {  
        String str1="Hello";  
        String str2="JLC";  
  
        System.out.println(str1+"\t"+str2);  
  
        String temp= str1;  
        str1=str2;  
        str2=temp;  
  
        System.out.println(str1+"\t"+str2);  
    }  
}
```

```
// Time Complexity - O(1)
```

```
//Space Complexity - O(1)
```



4) Lab4.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 * */
```

//4. Find Min and max of 2 Numbers

```
public class Lab4 {  
  
    public static void main(String[] args) {  
        int a=10;  
        int b=20;  
  
        int min = (a<b) ? a:b;  
        System.out.println(min);  
  
        int max=0;  
  
        if(a<b)  
            max=a;  
        else  
            max=b;  
  
        System.out.println(max);  
    }  
}
```

```
// Time Complexity - O(1)
```

```
//Space Complexity - O(1)
```



5) Lab5.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 * */
```

//5. Find max of 3 Numbers

```
class Lab5 {  
    public static void main(String as[]) {  
        int a=10;  
        int b=20;  
        int c=15;  
  
        int max=(a>b) ? ( (a>c) ? a : c) : ((b>c) ? b : c);  
        System.out.println(max);  
  
        if(a>b && a> c)  
            max=a;  
        else if(b> a && b>c)  
            max = b;  
        else  
            max=c;  
  
        System.out.println("Max value is "+max);  
    }  
}
```

// Time Complexity - O(1)
//Space Complexity - O(1)



6) Lab6.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//6. Print the Numbers from 1 to N
```

```
public class Lab6 {
```

```
    public static void main(String[] args) {  
        int n = 25;
```

```
        for (int i = 1; i <= n; i++) {  
            System.out.println(i);  
        }
```

```
    }
```

```
}
```

```
// Time Complexity - O(n)
```

```
//Space Complexity - O(1)
```



7) Lab7.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//7. Print the Even Numbers from 1 to N
```

```
public class Lab7 {
```

```
    public static void main(String[] args) {
```

```
        int n = 6;
```

```
        for (int i = 1; i <= n; i++) {
```

```
            if (i % 2 == 0)
```

```
                System.out.println(i);
```

```
        }
```

```
    }
```

```
}
```

```
// Time Complexity - O(n)
```

```
//Space Complexity - O(1)
```



8) Lab8.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//8. Find the Sum of Numbers from 1 to N
```

```
public class Lab8 {
```

```
    public static void main(String[] args) {  
        int n = 6;
```

```
        int sum=0;  
        for (int i = 1; i <= n; i++) {  
            sum = sum+i;  
        }
```

```
        System.out.println("Sum : "+sum);
```

```
    }  
}
```

```
// Time Complexity - O(n)
```

```
//Space Complexity - O(1)
```



9) Lab9.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//9. Find the Sum of Numbers from 1 to N
```

```
public class Lab9 {
```

```
    public static void main(String[] args) {  
        int n =5;
```

```
        int sum=n * (n+1)/2;  
        System.out.println("Sum : "+sum);
```

```
    }
```

```
}
```

```
// Time Complexity - O(1)
```

```
//Space Complexity - O(1)
```



10) Lab10.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//10. Find the Sum of Squares of Numbers from 1 to N
```

```
public class Lab10 {
```

```
    public static void main(String[] args) {  
        int n = 3;
```

```
        int sum=0;  
        for (int i = 1; i <= n; i++) {  
            sum = sum+i*i;  
        }
```

```
        System.out.println("Sum : "+sum);
```

```
    }
```

```
}
```

```
// Time Complexity - O(n)
```

```
//Space Complexity - O(1)
```




11) Lab11.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 * */
```

```
//11. Find the Sum of Cubes of Numbers from 1 to N
```

```
public class Lab11 {  
  
    public static void main(String[] args) {  
        int n = 5;  
  
        long sum=0;  
        for (int i = 1; i <= n; i++) {  
            sum = sum+i*i*i;  
        }  
  
        System.out.println("Sum : "+sum);  
    }  
}  
  
// Time Complexity - O(n)  
//Space Complexity - O(1)
```



12) Lab12.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//12. Print the Divisibles of 6 from 1 to N
```

```
public class Lab12 {  
    public static void main(String[] args) {  
        int n = 25;  
        for (int i = 1; i <= n; i++) {  
            if (i % 6 == 0)  
                System.out.println(i);  
        }  
    }  
}
```

```
// Time Complexity - O(n)
```

```
//Space Complexity - O(1)
```



13) Lab13.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
//13. Print the Divisibles of given Number
```

```
public class Lab13 {  
    public static void main(String[] args) {  
        int n = 10;  
  
        for (int i = 1; i <= n/2; i++) {  
            if (n % i == 0)  
                System.out.println(i);  
        }  
        System.out.println(n);  
    }  
}
```

```
// Time Complexity -  $O(n/2)$ 
```

```
//Space Complexity -  $O(1)$ 
```



14) Lab14.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
//14. Find the Sum of Divisibles of given Number(exclude N)
```

```
public class Lab14 {  
    public static void main(String[] args) {  
        int n = 28;  
  
        int sum = 0;  
        for (int i = 1; i <= n/2; i++) {  
            if (n % i == 0)  
                sum +=i;  
        }  
        System.out.println("Sum : "+ sum);  
    }  
}
```

```
// Time Complexity - O(n/2)
```

```
//Space Complexity - O(1)
```



15) Lab15.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
//15. Check whether Given Number is Prime or Not
```

```
public class Lab15 {
```

```
    public static boolean isPrime(int n) {  
        int count = 0;  
        for (int i = 2; i <= n / 2; i++) {  
            if (n % i == 0)  
                count++;  
        }  
    }
```

```
    if (count == 0)  
        return true;  
    else  
        return false;
```

```
    }  
  
    public static void main(String[] args) {  
        int n = 10;  
        boolean flag = isPrime(n);  
        System.out.println(flag);  
    }
```

```
}
```

```
// Time Complexity -  $O(n/2)$ 
```

```
//Space Complexity -  $O(1)$ 
```



16) Lab16.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//16. Check whether Given Number is Prime or Not
// Optimized

public class Lab16 {

    // Time Complexity - O(log n)
    // Time Complexity - O(n/2)
    public static boolean isPrime(int n) {
        int count = 0;
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                count++;
                break;
            }
        }

        if (count == 0)
            return true;
        else
            return false;
    }

    public static void main(String[] args) {
        int n = 100;
        boolean flag = isPrime(n);
        System.out.println(flag);
    }
}
```



17) Lab17.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//17. Check whether Given Number is Prime or Not
// More Optimized

public class Lab17 {

    // Time Complexity - O(log n)
    // Time Complexity - O(n/2)
    public static boolean isPrime(int n) {

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                return false;
            }
        }
        return true;
    }

    public static void main(String[] args) {
        int n = 10;
        boolean flag = isPrime(n);
        System.out.println(flag);
    }
}
```



18) Lab18.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//18. Print Prime Numbers from 1 to N

public class Lab18 {

    public static boolean isPrime(int n) {

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                return false;
            }
        }
        return true;
    }

    public static void main(String[] args) {
        int n = 50;
        for(int i=1;i<=n;i++) {
            if(isPrime(i))
                System.out.println(i);
        }
    }

}

// Time Complexity - O(n/2)
//Space Complexity - O(1)
```




19) Lab19.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

//14. Count the Divisibles of given Number

```
public class Lab19 {  
  
    public static void main(String[] args) {  
        int n = 10;  
  
        int count = 2;  
        for (int i = 2; i <= n/2; i++) {  
            if (n % i == 0)  
                count++;  
        }  
  
        System.out.println("Count : "+ count);  
    }  
}
```

// Time Complexity - $O(n/2)$

//Space Complexity - $O(1)$



20) Lab20.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//20. Check whether given number is Perfect Number or Not
```

```
public class Lab20 {  
  
    public static boolean isPerfect(int n) {  
        int sum = 0;  
        for (int i = 1; i <= n / 2; i++) {  
            if (n % i == 0)  
                sum += i;  
        }  
  
        if (sum == n)  
            return true;  
        else  
            return false;  
    }  
  
    public static void main(String[] args) {  
        int n = 60;  
        boolean flag = isPerfect(n);  
        System.out.println(flag);  
    }  
}
```

```
// Time Complexity -  $O(n/2)$ 
```

```
//Space Complexity -  $O(1)$ 
```



21) Lab21.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//21. Print Perfect Numbers from 1 to N

public class Lab21 {

    public static boolean isPerfect(int n) {
        int sum = 0;
        for (int i = 1; i <= n / 2; i++) {
            if (n % i == 0)
                sum += i;
        }

        if (sum == n)
            return true;
        else
            return false;
    }

    public static void main(String[] args) {
        int n = 10000;
        for(int i=1;i<=n;i++) {
            if(isPerfect(i))
                System.out.println(i);
        }
    }

    // Time Complexity -  $O(n * n/2)$  /  $O(n^2)$ 
    //Space Complexity -  $O(1)$ 
```



22) Lab22.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//22. Print the Digits of Given Number

public class Lab22 {

    public static void main(String[] args) {
        int n = 12345;

        while(n!=0) {
            int digit = n%10;
            System.out.println(digit);
            n= n/10;
        }
    }

}

// Time Complexity - O(log n)
//Space Complexity - O(1)
```



23) Lab23java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//23. Count the Digits of Given Number
```

```
public class Lab23 {  
  
    public static void main(String[] args) {  
        int n = 12345;  
  
        int count=0;  
        while(n!=0) {  
            //int digit = n%10;  
            count++;  
            n= n/10;  
        }  
        System.out.println(count);  
    }  
}
```

```
// Time Complexity -  $O(\log n)$ 
```

```
//Space Complexity -  $O(1)$ 
```



24) Lab24.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 * */
```

```
//24. Sum the Digits of Given Number
```

```
public class Lab24 {  
  
    public static void main(String[] args) {  
        int n = 123;  
  
        int sum=0;  
        while(n!=0) {  
            int digit = n%10;  
            sum = sum +digit;  
            n= n/10;  
        }  
        System.out.println(sum);  
    }  
}
```

```
// Time Complexity -  $O(\log n)$ 
```

```
//Space Complexity -  $O(1)$ 
```



25) Lab25.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//25. Sum the Squares of Digits of Given Number
```

```
public class Lab25 {
```

```
    public static void main(String[] args) {
```

```
        int n = 123;
```

```
        int sum=0;
```

```
        while(n!=0) {
```

```
            int digit = n%10;
```

```
            sum = sum +digit * digit;
```

```
            n= n/10;
```

```
        }
```

```
        System.out.println(sum);
```

```
    }
```

```
}
```

```
// Time Complexity -  $O(\log n)$ 
```

```
//Space Complexity -  $O(1)$ 
```



26) Lab26.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//26. Sum the Cubes of Digits of Given Number
```

```
public class Lab26 {  
  
    public static void main(String[] args) {  
        int n = 123;  
  
        int sum=0;  
        while(n!=0) {  
            int digit = n%10;  
            sum = sum +digit * digit * digit;  
            n= n/10;  
        }  
        System.out.println(sum);  
    }  
}
```

```
// Time Complexity -  $O(\log n)$ 
```

```
//Space Complexity -  $O(1)$ 
```




27) Lab27.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
//27. Check whether given number is Arm Strong Number or Not
```

```
public class Lab27 {  
    public static boolean isArmStrong(int n) {  
  
        int nCopy=n;  
        int sum = 0;  
        while (n != 0) {  
            int digit = n % 10;  
            sum = sum + digit * digit * digit;  
            n = n / 10;  
        }  
  
        if (sum ==nCopy)  
            return true;  
        else  
            return false;  
    }  
  
    public static void main(String[] args) {  
        int n = 372;  
        boolean flag = isArmStrong(n);  
        System.out.println(flag);  
    }  
}
```

```
// Time Complexity -  $O(\log n)$ 
```

```
//Space Complexity -  $O(1)$ 
```



28) Lab28.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//28. Print Arm Strong Numbers from 1 to N

public class Lab28 {
    public static boolean isArmStrong(int n) {

        int nCopy = n;
        int sum = 0;
        while (n != 0) {
            int digit = n % 10;
            sum = sum + digit * digit * digit;
            n = n / 10;
        }
        if (sum == nCopy)
            return true;
        else
            return false;
    }

    public static void main(String[] args) {
        int n = 1000;
        for (int i = 1; i <= n; i++) {
            if (isArmStrong(i))
                System.out.println(i);
        }
    }
}

// Time Complexity - O(n log n)
//Space Complexity - O(1)
```



29) Lab29.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//29. Find the Factorial of Given Number
```

```
public class Lab29 {
```

```
    public static void main(String[] args) {  
        int n = 20;
```

```
        long fact = 1;  
        for (int i = 1; i <= n; i++){  
            fact = fact * i;  
        }
```

```
        System.out.println(fact);
```

```
    }
```

```
}
```

```
// Time Complexity - O(n)
```

```
//Space Complexity - O(1)
```



30) Lab30.java

```
package com.myjlc.loops;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
//30. Check whether given number is Strong Number or Not.
public class Lab30 {
    public static long factorial(int n) {
        if(n==0 || n==1)
            return 1;

        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }
    public static boolean isStrong(int n) {
        int nCopy = n;

        long sum = 0;
        while (n != 0) {
            int digit = n % 10;
            long fact = factorial(digit);
            sum += fact;
            n = n / 10;
        }
        return (sum == nCopy);
    }
    public static void main(String[] args) {
        int n = 100;
        boolean flag = isStrong(n);
        System.out.println(flag);
    }
}
// Time Complexity - O(log n)
//Space Complexity - O(1)
```



31) Lab31.java

```
package com.jlcindia;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
//31. Print the Strong Numbers from 1 to N
public class Lab31 {
    public static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }
    public static boolean isStrong(int n) {
        int nCopy = n;

        long sum = 0;
        while (n != 0) {
            int digit = n % 10;
            long fact = factorial(digit);
            sum += fact;
            n = n / 10;
        }
        return (sum == nCopy);
    }
    public static void main(String[] args) {
        int n = 1000;

        for(int i=1;i<=n;i++) {
            if(isStrong(i))
                System.out.println(i);
        }
    }
}
// Time Complexity - O(n log n)
//Space Complexity - O(1)
```



32) Lab32.java

```
package com.myjlc.loops;
```

```
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */
```

```
//32. Print the Reverse of given Number
```

```
public class Lab32 {  
  
    public static void main(String[] args) {  
        int n = 12345;  
  
        int rev = 0;  
  
        while (n != 0) {  
            rev = rev * 10 + n % 10;  
            n = n / 10;  
        }  
        System.out.println(rev);  
    }  
}
```

```
// Time Complexity - O(d) where d- No of Digits
```

```
//Space Complexity - O(1)
```



33) Lab33.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//33. Check whether given number is Palindrome or Not
```

```
public class Lab33 {
```

```
    public static boolean isPalindrome(int n) {
```

```
        int nCopy=n;
```

```
        int rev = 0;
```

```
        while (n != 0) {
```

```
            rev = rev * 10 + n % 10;
```

```
            n = n / 10;
```

```
        }
```

```
        System.out.println(rev);
```

```
        return (rev==nCopy);
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        int n = 121;
```

```
        boolean flag=isPalindrome(n);
```

```
        System.out.println(flag);
```

```
    }
```

```
}
```

```
// Time Complexity - O(d) where d- No of Digits
```

```
//Space Complexity - O(1)
```



34) Lab34.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
*/
```

```
//34. Print the Digits of Given Number in same Order
```

```
// n= 123 - 1, 2, 3
```

```
public class Lab34 {  
    public static void main(String[] args) {  
        int n = 12345;  
  
        int rev = 0;  
  
        while (n != 0) {  
            int digit = n % 10;  
            rev = rev * 10 + digit;  
            n = n / 10;  
        }  
  
        while (rev != 0) {  
            int digit = rev % 10;  
            System.out.println( digit % 10);  
            rev = rev / 10;  
        }  
    }  
}
```

```
// Time Complexity - O(d) where d- No of Digits
```

```
//Space Complexity - O(1)
```




35) Lab35.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

// 29) WAP to print the fibonocii series.
// 0 1 1 2 3 5 8 13

public class Lab35 {
    public static void main(String[] args) {

        int n =10;

        int a = 0;
        int b = 1;

        System.out.print(a+"\t"+b+"\t");

        for(int i=1;i<=n-2;i++) {
            int next = a+b;
            System.out.print(next+"\t");
            a=b;
            b=next;
        }
    }
}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



36) Lab36.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//36. Print the following Series
//1,1,6,4,11,9,16,16,21,25

public class Lab36 {

    public static void main(String[] args) {

        int n = 5;

        int a = 1;
        int b=1;

        for (int i = 1; i <= n; i++) {
            //1. First Series
            System.out.print(a + "\t");
            a = a + 5;

            //2. Second Series
            b= i*i;
            System.out.print(b+"\t");
        }
    }

}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



37) Lab37.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

//37. Print the following Series
//0, 1, 3, 8, 6, 27, 9, 64, 12, 125

public class Lab37 {
    public static void main(String[] args) {

        int n = 5;

        int a = 0;
        int b=1;

        for (int i = 1; i <= n; i++) {
            //1. First Series
            System.out.print(a + "\t");
            a = a + 3;

            //2. Second Series
            b= i*i*i;
            System.out.print(b+"\t");
        }
    }

}

// Time Complexity - O(n)
//Space Complexity - O(1)
```



38) Lab38.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//38. Evaluate the following Services
```

```
// 1 + 2/x+3/x2+4/x3+....
```

```
public class Lab38 {  
    public static double power(double x,double p) {  
        double result=1;  
        for(int i=1;i<=p;i++) {  
            result= result*x;  
        }  
        return result;  
    }  
}
```

```
    public static void main(String[] args) {  
  
        int n = 5;  
        int x=2;  
  
        double result = 0.0;  
        for (int i = 1; i <= n; i++) {  
            result = result + i/power(x,i-1);  
        }  
        System.out.println(result);  
    }  
}
```

```
// Time Complexity - O(n)/O(n*p)
```

```
//Space Complexity - O(1)
```



39) Lab39.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//39. Evaluate the following Services
```

```
// 1 - 2/x+3/x2-4/x3+....
```

```
public class Lab39 {  
    public static double power(double x,double p) {  
        double result=1;  
        for(int i=1;i<=p;i++) {  
            result= result*x;  
        }  
        return result;  
    }  
    public static void main(String[] args) {  
  
        int n = 5;  
        int x=2;  
        double result = 0.0;  
        int sign=1;  
        for (int i = 1; i <= n; i++) {  
            result = result + (sign)*( i/power(x,i-1));  
            sign = sign*-1;  
        }  
        System.out.println(result);  
    }  
}
```

```
//Time Complexity - O(n)/O(n*p)
```

```
//Space Complexity - O(1)
```



40) Lab40.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//40. Evaluate the following Services
```

```
//  $1/(x)! + x/(x+1)! + x^2/(x+2)! + x^3/(x+3)! + \dots$ 
```

```
public class Lab40{
```

```
}
```

```
//Time Complexity -
```

```
//Space Complexity -
```



41) Lab41.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
//41. Evaluate the following Services
```

```
//  $1/(x)! - x/(x+1)! + x^2/(x+2)! - x^3/(x+3)! + \dots$ 
```

```
public class Lab41{
```

```
}
```

```
//Time Complexity -
```

```
//Space Complexity -
```



42) Lab42.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
/*
42) Print the following Pattern
*****
*****
*****
*****
*****

*/
public class Lab42 {
    public static void main(String[] args) {

        int rows = 6;
        int stars = 6;

        for (int i = 1; i <= rows; i++) {
            // 1. Print Stars
            for (int k = 1; k <= stars; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");
        }
    }
}

//Time Complexity - O(rows * stars)
//Space Complexity - O(1)
```




43) Lab43.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
/*
```

43) Print the following Pattern

```
  *
 * *
* * *
* * * *
* * * * *
```

```
*/
```

```
public class Lab43 {
    public static void main(String[] args) {

        int rows = 6;
        int stars = 1;

        for (int i = 1; i <= rows; i++) {
            // 1. Print Stars
            for (int k = 1; k <= stars; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");

            stars=stars+1;
        }
    }
}
```

```
//Time Complexity - O(rows * stars)
```

```
//Space Complexity - O(1)
```



44) Lab44.java

```
package com.myjlc.loops;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @Company: Java Learning Center
```

```
* */
```

```
/*
```

44) Print the following Pattern

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

```
*/
```

```
public class Lab44 {
    public static void main(String[] args) {

        int rows = 7;
        int stars = 1;

        for (int i = 1; i <= rows/2+1; i++) {
            // 1. Print Stars
            for (int k = 1; k <= stars; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");

            stars=stars+1;
        }
    }
}
```



```
        for (int i = 1; i <= rows/2; i++) {

            // 1. Print Stars
            for (int k = 1; k <= stars-2; k++) {
                System.out.print(" * ");
            }
            // 2. Print New Line
            System.out.println("");

            stars=stars-1;

        }

    }

}

//Time Complexity - O(rows * stars)
//Space Complexity - O(1)
```

45) Lab45.java

```
package com.myjlc.loops;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

/*
45) Print the following Pattern
    *
   ***
  *****
 *****

 */
```



```
public class Lab45 {  
    public static void main(String[] args) {  
  
        int rows = 10;  
        int spaces=25;  
        int stars = 1;  
  
        for (int i = 1; i <= rows; i++) {  
            //1.Print Spaces  
            for (int k = 1; k <= spaces; k++) {  
                System.out.print(" ");  
            }  
  
            // 2. Print Stars  
            for (int k = 1; k <= stars; k++) {  
                System.out.print(" * ");  
            }  
  
            // 3. Print New Line  
            System.out.println("");  
  
            spaces = spaces - 2;  
            stars=stars+2;  
        }  
    }  
}  
  
//Time Complexity - O(rows * stars)  
//Space Complexity - O(1)
```



46) Lab46.java

```
package com.jlcindia;  
/*  
* @Author : Srinivas Dande  
* @Company: Java Learning Center  
**/  
/*
```

46) Print the following Pattern

```
      *  
    * * *  
  * * * * *  
* * * * * * *  
  * * * * *  
    * * *  
      *
```

```
*/  
public class Lab46 {  
    public static void main(String[] args) {  
        int rows = 11;  
        int spaces=25;  
        int stars = 1;  
  
        for (int i = 1; i <= rows/2; i++) {  
            //1.Print Spaces  
            for (int k = 1; k <= spaces; k++) {  
                System.out.print(" ");  
            }  
  
            // 2. Print Stars  
            for (int k = 1; k <= stars; k++) {  
                System.out.print(" * ");  
            }  
  
            // 3. Print New Line  
            System.out.println("");  
  
            spaces = spaces - 2;  
            stars=stars+2;  
        }  
    }  
}
```



```
for (int i = 1; i <= rows/2+1; i++) {  
    //1.Print Spaces  
    for (int k = 1; k <= spaces; k++) {  
        System.out.print(" ");  
    }  
  
    // 2. Print Stars  
    for (int k = 1; k <= stars; k++) {  
        System.out.print(" * ");  
    }  
  
    // 3. Print New Line  
    System.out.println("");  
  
    spaces = spaces + 2;  
    stars=stars-2;  
}  
}  
}  
  
//Time Complexity - O(rows * stars)  
//Space Complexity - O(1)
```